

Tensor computations

Improve the worst-case approximation factor for prescribed tensor-train ranks

Difficulty: extreme **Importance:** interesting to the community**Status:** Open

Rating rationale: Extreme because the guarantee must improve a general worst-case approximation barrier without increasing ranks; community importance reflects the central role of TT compression in numerical tensor methods.

Problem statement

Let $d \geq 3$, $n_1, \dots, n_d \geq 2$, and positive integers $r_1, \dots, r_{d-1}$ be given. Let $S_r \subset \mathbb{R}^{n_1 \times \dots \times n_d}$ consist of tensors whose unfolding separating modes $1, \dots, j$ from modes $j+1, \dots, d$ has rank at most r_j , for every j . For a dense input tensor A , put

$$E_*(A, r) = \min_{Y \in S_r} \|A - Y\|_F^2.$$

Find a polynomial-time algorithm returning $X \in S_r$ with

$$\|A - X\|_F^2 < (d-1)E_*(A, r) \quad \text{whenever } E_*(A, r) > 0,$$

and exact reconstruction when $E_*(A, r) = 0$; alternatively, establish a complexity obstruction to such a guarantee. Ranks may not be increased. Polynomial time is measured in the dense input size and rank parameters, in the idealized arithmetic/SVD model used for TT-SVD; a bit-complexity formulation must additionally specify precision and output tolerances.

This is the tensor-train case of the published tree-network question. The positive-error qualification corrects the impossible strict inequality at zero optimum. It asks for a uniformly valid algorithm, not an empirical improvement or a guarantee restricted to particular input tensors.

References

I. V. Oseledets, *Tensor-Train Decomposition*, *SIAM Journal on Scientific Computing* 33 (2011), 2295–2317, Theorem 2.2 and Corollary 2.4. Amsel et al., [workshop report](#), §6.1, Problem 6.1. Matthew Fahrbach and Mehrdad Ghadiri, *A Tight Lower Bound for the Approximation Guarantee of Higher-Order Singular Value Decomposition*, Theorems 1.2–1.3, concerns tightness of specific Tucker algorithms rather than a lower bound against all tensor-train algorithms.

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Rechecked [workshop v3](#), §6.1, Problem 6.1, which still requests a strict improvement or a complexity obstruction. Searches for tensor-train approximation improvements in 2026 also located [Yu et al., randomized block Krylov TT approximation](#): its probabilistic error estimates and practical speedups do not give the displayed universal improvement at unchanged ranks. No full resolution was located; the positive-optimum qualification remains essential.